



Mastering CTF Tools and Attack Strategies

Document Identification Table	
Title	Mastering CTF Tools and Attack Strategies
Version	1.0.0
Creation date	08/2026
Last updated	08/2026
Author	0xm4rk1h
Status	Released
Classification	Public

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1 MAIN INFORMATION

1.1 Warning

The information and tools presented are intended for educational purposes only, to raise awareness and provide instruction. I would like to remind you that any use of the knowledge acquired must comply with the laws in force. The use of this knowledge for illegal or malicious activities is strictly forbidden and may result in legal consequences. I would like to make it clear that I cannot be held responsible for any illegal acts or behavior that may result from the use of this knowledge.

1.2 Course aim

Set up an attack strategy to make effective use of the tools for a successful CTF challenge.

1.3 Course requirements

- Basic Linux knowledge.
- Basic networking knowledge.
- Basic web knowledge.
- Basic cybersecurity knowledge.

2 ATTACK STRATEGY

2.1 Introduction

To successfully complete CTF challenges, I have developed an attack strategy that divides the process into three main phases. This strategy is inspired by the **Cyber Kill Chain** and provides a structured approach to analyzing and attacking a target. The three phases of this attack strategy are:

2.2 Phase 1 - Reconnaissance

The objective of this phase is to gather information about the target, such as the technologies in use, open ports, available services, and potential vulnerabilities or weaknesses. The more information we gather during this phase, the better we can understand the target and identify potential attack vectors.

2.3 Phase 2 - Exploitation

This phase focuses on exploiting the vulnerabilities identified during the reconnaissance phase. The objective is to gain initial access to the target system.

2.4 Phase 3 - Total Control & Evasion

Once initial access has been obtained, the goal is to stabilize and extend our access. This may involve escalating privileges, moving from a restricted access level to a higher level of privileges, and identifying alternative ways to maintain access. Once the objectives of the challenge have been achieved, the final step is to clean up the environment and remove any changes made during the exercise.

Note

All techniques presented in this article are intended for use in authorized CTF challenges and controlled lab environments.

3 TOOLS

3.1 Introduction

For each phase of the attack strategy, different tools and online resources can be used to analyze, exploit, and understand the target. The following selection introduces some of the tools and websites commonly used in CTF challenges.

3.2 Phase 1 - Reconnaissance

- **Nmap**: Network exploration and port scanning.
- **Wireshark**: Network packet analyzer.
- **Gobuster**: Directory and file enumeration on web servers.
- **FFUF**: Web enumeration, directory brute-forcing, and fuzzing.
- **Enum4linux**: Enumeration of SMB information, such as shares and users, on Windows and Linux systems.
- **Exploit-DB**: Archive of publicly available exploits and information about vulnerable software.
- **MITRE ATT&CK**: Knowledge base of tactics and techniques used to understand, classify, and analyze cyber attacks.

3.3 Phase 2 - Exploitation

- **Burp Suite**: Security testing of web applications.
- **Hydra**: Password and login brute-forcing tool supporting multiple protocols.
- **John the Ripper**: Password hash cracking tool.
- **Hashcat**: High-performance password hash recovery tool.
- **CrackStation**: Password hash cracking.
- **Metasploit**: Penetration testing framework.
- **SQLMap**: Detecting and exploiting SQL injection vulnerabilities.
- **Reverse Shell Generator**: Generating reverse shell commands.
- **Reverse Shell Cheat Sheet**: Reference for commonly used reverse shell commands.
- **Spawn Shell**: More stable interactive shells.

3.4 Phase 3 - Total Control & Evasion

- **LinPEAS / WinPEAS:** Tools for identifying potential privilege escalation paths on Linux/Unix and Windows systems.
- **GTFOBins:** A curated list of Unix binaries that can be abused to bypass local security restrictions in misconfigured systems.

3.5 Miscellaneous

- **CyberChef:** Web-based tool for encoding, decoding, transforming, and analyzing data.

This is only a short list of the tools and resources that can be useful during CTF challenges. There are, of course, many other tools, techniques, and resources that can be used depending on the target and the type of challenge. However, this selection provides a good starting point for beginners who want to develop their CTF skills and become familiar with commonly used tools.

4 THANKS

Keep learning and you'll reach your goals.

Thank you for reading this content.

See you soon.

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